

Design Document: job-crawler

Authored by Mason Hart and Tyler Mestery

Introduction:

A basic agent-driven app that scrapes popular job sites (linkedin, indeed, etc.) 3-6 times daily to find job postings that match a users' preferences (resume, role title, etc.) and organizes them into a list format on our local-hosted server website.

Purpose:

The purpose of this app is to make a centralized and easy experience for quickly searching and applying to jobs. However, in order to make informed decisions, clients will need lots of information about the job position in order to apply. As such, the design should be as simple as possible on its face, but users should be able to find more information easily upon clicking. Overall, our purpose is to provide a very useful service for consolidating all relevant job postings in one place, instead of having to scour through multiple websites such as indeed, linkedin, and company websites for roles. We want our clients to be able to personalize the experience and their feed to the career path that they describe. We should stress again that this app should be simple and easy to use and parse, but should still contain as much necessary information about the role as possible.

Requirements:

Nonfunctional

- Runs the scraper 3-6 times daily to find new job postings
- Only displays job postings that align with user's resume and/or preferences
- Has a note that appears when another user already applied for a job posting that appears on their feed
- User is able to mark postings as applied
- User is able to remove job postings from feed
- User interface is clean and easy to glance through
- Profiles page contains a resume upload
- Users can see other users job postings they applied to by looking at the other users profile page
- There is a total applied counter on each users' profile page

- Users can filter open job postings displayed to them with a search bar
- Can also filter through job postings by newest, company, title, location, and money
- Users can opt in to receive email notifications (in their profile page/on signup) for that days new job postings by a job posting bot daily
- Web parsing should be fast

Functional

- Users can log in and sign up
- Users can set and update their job preferences
- The app emails users frequently about new job openings
- Users can view new job openings
- Users can view job openings they and others have applied to

Tech Stack:

React (TS + CSS) handles the dynamic user interface and state in the browser, while Golang manages server-side logic, database interactions, and API endpoints.

Frontend: Typescript + CSS + React

Backend: Golang

Database: MongoDB (local to server, not cloud-hosted)

Concept Sketches:

- Located in /documents
- Two images: one for representation of homepage and another of profilepage

Resources and Tools:

- Utilize an API to send email notifications
- Provide clear documentation & readme
- Provide easy setup instructions for our server's MongoDB when set up there

Features In Detail:

User authentication

- There should be a signup page that accepts an email, username, and password. This page should lead to another that asks for basic user preferences, which will be used to determine what job openings the web crawler should look for to give to this user
- There should be a login page with a form that accepts an email and password. This should sign in the user and bring them directly to their feed.

Job openings feed

- This is the main page of the app. It shows all of the basic information about a given job opening, and it shows a list of these. It only shows postings that the user has not applied to. Each entry should have buttons for 'accept' and 'reject'.
 - On 'accept', the user is indicating that they have applied to the position. The entry is saved in the database and removed from the homepage. The user will be able to see this in their profile/history page.
 - On 'reject', the job opening is removed from the home page and not saved. The user should never see this opening again

Feed personalization

- There should be a profile page. On this page, if the user is on their own profile page, the user should be able to set their preferences such as adding their resume and whether they should be receiving emails. Additionally, there should be the list of job openings that the user has indicated that they have applied to. This may be a big list over time, so loading speed should be accounted for by the frontend. This history should be available to everyone, even different profiles should have their history available on their profile page, UNLESS THE ACCOUNT IS PRIVATE. This means that there should be a spot for users to go to the profile pages of other users in the homepage.
- Allow for searching (search by words, i.e. software engineer roles minnesota)
- Allow for filtering by compensation, company, job title, newest, location
- Add a light or dark mode option

Application History:

- See past applications that have been marked as "applied" by the user
- See the date that the user marked that job posting as "applied"

- See which friends (if any) on the platform, with accounts, also applied to that posting

Final Comments:

- Focus on writing maintainable code
- Provide clear documentation - don't write unreadable code
- Utilize comments clearly and effectively - no unnecessary emojis for example 😊
- Put effort into the security of the website, focus on writing code without exploits